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Inventor(s)	Jeannin; Olivier et al.

Optoelectronic device with micrometric or nanometric light-emitting diode on which an optical lens is mounted

Abstract

An optoelectronic device includes at least one light-emitting diode having a three-dimensional shape having a height along a longitudinal axis and having a first longitudinal dimension measured along the longitudinal axis and at least a second transverse dimension corresponding to a dimension of the three-dimensional shape measured perpendicular to the longitudinal axis. The first longitudinal dimension and the second transverse dimension are each less than or equal to substantially 20 μm. The optoelectronic device has at least one optical lens capable of transforming the light rays emitted by the light-emitting diode which pass through the optical lens.

Inventors:	**Jeannin; Olivier (Grenoble, FR), Mayer; Frédéric (Voiron, FR)**
Applicant:	**ALEDIA** (Échirolles, FR)
Family ID:	**66867330**
Assignee:	**ALEDIA (Echirolles, FR)**
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Primary Examiner: Patel; Reema

Attorney, Agent or Firm: CANTOR COLBURN LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure concerns an optoelectronic device including a substrate delimiting a support face and at least one light-emitting diode formed on the support face, the light-emitting diode having a three-dimensional shape including a height along a longitudinal axis extending according to a first direction directed transversely to the support face.

(2) The disclosure also concerns a method for manufacturing an optoelectronic device.

(3) The disclosure finds application in particular in display screens or images projection systems.

BACKGROUND

(4) By optoelectronic device, it should be understood herein a device adapted to perform the conversion of an electrical signal into an electromagnetic radiation to be emitted, in particular light.

(5) There are optoelectronic devices including light-emitting diodes, also known under the acronym LED, formed on a substrate.

(6) It is known that each light-emitting diode comprises an active material exploiting quantum wells, or not, a semiconductor portion doped according to a first doping type to serve as a P-doped junction and a semiconductor portion doped according to a second doping type to serve as a N-doped junction.

(7) Each light-emitting diode may be made based on micrometric or nanometric semiconductor three-dimensional elements, which, in turn, are at least partially obtained by epitaxial growth such as by metal organic chemical vapor deposition (MOCVD) or by molecular beam epitaxy (MBE) or by hydride vapor phase epitaxy (HYPE). Typically, the light-emitting diodes are made based on a semiconductor material comprising for example elements from the column III and from the column V of the periodic table, such as a III-V compound, in particular gallium nitride (GaN), indium and gallium nitride (InGaN) or aluminum and gallium nitride (AlGaN).

(8) There are optoelectronic devices including an array of light-emitting diodes having a determined emission surface throughout which is transmitted the light radiation emitted by the light-emitting diodes. In particular, such optoelectronic devices may be used in the making of display screens or images projection systems, where the array of light-emitting diodes actually defines an array of light pixels where each pixel conventionally includes at least one sub-pixel for generating each color, each sub-pixel containing, in turn, at least one light-emitting diode. For example, a sub-pixel may contain up to 100000 light-emitting diodes.

(9) One of the difficulties is that as the definition of screens increases, the dimensions between the light-emitting diodes and the dimensions of each sub-pixel become micrometric, or nanometric, and the use of three-dimensional light-emitting diodes becomes inevitable with increasingly smaller dimensions of three-dimensional light-emitting diodes. Yet, the luminous intensity emitted by the nanometric three-dimensional light-emitting diodes declines drastically as the size of the three-dimensional light-emitting diodes is reduced.

(10) Another difficulty is how to ensure that the light radiation emitted by the three-dimensional light-emitting diodes of a determined sub-pixel does not mix with the light radiation emitted by the light-emitting diodes of an adjacent sub-pixel in order to improve the contrasts. Yet, this problem turns out to be more and more difficult given the increasing miniaturization of light-emitting diodes.

(11) Another difficulty arises from the fact that the light emitted by a three-dimensional diode is not directional. This results in a considerable radiation loss by emission in undesirable directions, thereby causing a drop in luminous efficacy and in overall efficiency.

(12) A known solution consists in forming light confinement walls adapted to block the transmission of the light radiation emitted by at least one determined light-emitting diode towards

at least one adjacent light-emitting diode.
(13) But this known technique does not address the need for making the light rays directional or focused and this further results in a partial loss of the light radiation by absorption through the light confinement walls, making the luminous efficacy and the overall efficiency drop.

SUMMARY

(14) The present disclosure aims to address all or part of the problems set out hereinbefore.
(15) In particular, to the disclosure provides a solution addressing at least one of the following objectives: focus or make the light rays emitted by a three-dimensional light-emitting diode as much directional as possible; increase the luminous efficacy of the optoelectronic device; allow obtaining the highest possible luminous intensity emitted by a three-dimensional light-emitting diode.
(16) This advantage can be achieved by the provision of an optoelectronic device including a substrate delimiting a support face and at least one light-emitting diode formed on the support face, said at least one light-emitting diode having a three-dimensional shape comprising a height according to a longitudinal axis extending according to a first direction directed transversely to the support face and having a first longitudinal dimension considered according to the longitudinal axis between a proximal portion of the light-emitting diode directed towards the support face and a distal portion of the light-emitting diode opposite, along the first direction, to said proximal portion, and at least one second transverse dimension corresponding to a dimension of said three-dimensional shape considered perpendicularly to the longitudinal axis, wherein each of the first longitudinal dimension and the second transverse dimension is smaller than or equal to about 20 μm, the optoelectronic device includes at least one optical lens adapted to transform the light rays emitted by the light-emitting diode that cross said optical lens, formed above the distal portion of the light-emitting diode according to the first direction in a manner generally axially aligned with the longitudinal axis of the light-emitting diode and having an outer surface adapted to be crossed by at least one portion of said light rays having a convex shape.
(17) Some preferred, yet non-limiting, aspects of the device are as follows.
(18) The convex shape of the outer surface of the optical lens is in the form of a surface of revolution whose axis of revolution is substantially aligned with the longitudinal axis of the light-emitting diode and whose generatrix is an arcuate segment.
(19) The surface of revolution has a section, viewed in any sectional plane including the axis of revolution, shaped like a horseshoe arch.
(20) The surface of revolution is shaped so that the surface area of the section of the outer surface in a section perpendicularly to its axis of revolution has a maximum value at the level of a main plane substantially parallel to the support face where the section has a maximum diameter larger than or equal to the second transverse dimension of the light-emitting diode above which the optical lens having said maximum diameter is formed.
(21) The surface of revolution is shaped so that the surface area of the section of the outer surface in a section perpendicularly to its axis of revolution has a maximum value at the level of a main plane substantially parallel to the support face where the section has a maximum diameter larger than or equal to the second transverse dimension of the light-emitting diode above which the optical lens having said maximum diameter is formed.
(22) The maximum diameter is comprised between 0.1 μm and 20 μm.
(23) The optical lens is made of a material having an optical index comprised between 1.4 and 2.2.
(24) On the side opposite to the distal portion of the light-emitting diode according to the first direction, the optical lens is truncated according to a plane substantially parallel to the support face.
(25) At least one portion of the light-emitting diode comprises a passivation layer.
(26) The optical lens and the distal portion of the light-emitting diode are in mechanical contact.
(27) At least one portion of the optical lens and the distal portion of the light-emitting diode are separated from one another with the interposition of a distance comprised between 5 nm and 20

μm.

(28) An optical material, with an optical index comprised between 1.4 and 2.2 and lower than or equal to an optical index of the optical lens, fills all or part of the space comprised between the distal portion of the light-emitting diode and the portion of the optical lens turned towards the distal portion of the light-emitting diode.

(29) All or part of the light-emitting diode is surrounded by an encapsulation material.

(30) When viewed perpendicularly to the longitudinal axis, the encapsulation material is surrounded by at least one wall adapted to reflect the light rays emitted by the light-emitting diode.

(31) The disclosure also covers the implementation of a method for manufacturing an optoelectronic device, the method comprising the following steps: a) formation of a substrate delimiting a support face; b) formation of at least one light-emitting diode having a three-dimensional shape comprising a height according to a longitudinal axis extending according to a first direction directed transversely to the support face and having a first longitudinal dimension considered according to the longitudinal axis between a proximal portion of the light-emitting diode directed towards the support face and a distal portion of the light-emitting diode opposite, along the first direction, to said proximal portion, and at least one second transverse dimension corresponding to a dimension of said three-dimensional shape considered perpendicularly to the longitudinal axis, wherein each of the first longitudinal dimension and the second transverse dimension is smaller than or equal to substantially 20 μm; i) formation of at least one optical lens above the distal portion of the light-emitting diode according to the first direction so that the optical lens is generally axially aligned with the longitudinal axis of the light-emitting diode, the formed optical lens being adapted to transform the light rays emitted by the light-emitting diode that cross the optical lens and having an outer surface adapted to be crossed by said light rays having a convex shape.

(32) Some preferred, yet non-limiting, aspects of the method are as follows.

(33) The manufacturing method comprises the following steps, implemented between step b) and step i): c) formation of an encapsulation material surrounding all or part of the light-emitting diode; d) formation, over the exposed face of the intermediate structure obtained at step c), of a first layer of an optical material, with an optical index comprised between 1.4 and 2.2 and lower than or equal to an optical index of the optical lens, said first optical material layer being adapted to be etched isotropically through the implementation of a first etching method; e) formation, over the free surface of the intermediate structure obtained after step d), of a hard mask layer adapted to not be etched during the use of the first etching method; f) etching of a first opening, through the implementation of a second etching method, through the hard mask layer obtained at step e), the first opening corresponding to a section of the light-emitting diode according to a plane parallel to the support face, said first opening of the hard mask layer being located above the light-emitting diode in the continuation of the longitudinal axis; g) etching of at least one cavity, through the implementation of the first etching method at the level of the first opening obtained at step f), across all or part of the thickness of the first optical material layer, the cavity obtained in this manner having a concave shape complementary to the convex surface of the optical lens and being suited for the subsequent formation of the optical lens at step i) directly in the cavity; h) removal of the hard mask layer.

(34) Step i) may comprise the following steps: i1) formation of a polymer layer over the exposed surface of the intermediate structure obtained after step h), the polymer layer filling all or part of the cavity; i2) etching of the polymer layer located at the level of at least one portion to be removed, allowing preserving at least one residual portion of the polymer layer located right above the light-emitting diode in the continuation of the longitudinal axis; i3) annealing of said at least one residual portion, annealing being carried out in conditions allowing making the polymer constituting said at least one residual portion pass through a viscous state.

(35) The manufacturing method may include a step j) of forming at least one wall adapted to reflect

the light radiations emitted by the light-emitting diode, the wall extending from the support face substantially parallel to the longitudinal axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other aspects, objects, advantages and features of the disclosure will appear better on reading the following detailed description of preferred embodiments thereof, provided as a non-limiting example, and made with reference to the appended drawings in which:

(2) FIG. **1** represents a schematic cross-section of a light-emitting diode adapted to be used in the disclosure;

(3) FIG. **2** represents a schematic cross-section of a first embodiment of an optoelectronic device according to the disclosure containing a light-emitting diode;

(4) FIG. **3** illustrates a schematic cross-section of a second embodiment of an optoelectronic device according to the disclosure containing a light-emitting diode;

(5) FIG. **4** represents a schematic cross-section of a third embodiment of an optoelectronic device according to the disclosure containing a light-emitting diode;

(6) FIG. **5** illustrates a schematic cross-section of a fourth embodiment of an optoelectronic device according to the disclosure containing a light-emitting diode;

(7) FIG. **6** represents, in a schematic cross-section, a first step of a first example of a method for manufacturing an optoelectronic device according to the disclosure;

(8) FIG. **7** represents, in a schematic cross-section, a second step of the first example of the manufacturing method;

(9) FIG. **8** represents, in a schematic cross-section, a third step of the first example of the manufacturing method;

(10) FIG. **9** represents, in a schematic cross-section, a fourth step of the first example of the manufacturing method;

(11) FIG. **10** represents, in a schematic cross-section, a fifth step of the first example of the manufacturing method;

(12) FIG. **11** represents, in a schematic cross-section, a sixth step of the first example of the manufacturing method;

(13) FIG. **12** represents, in a schematic cross-section, a first step of a second example of a method for manufacturing an optoelectronic device according to the disclosure;

(14) FIG. **13** represents, in a schematic cross-section, a second step of the second example of the manufacturing method;

(15) FIG. **14** represents, in a schematic cross-section, a seventh step of the first example of the manufacturing method; and

(16) FIG. **15** represents, in a schematic cross-section, a step of a third example of a method for manufacturing an optoelectronic device according to the disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

(17) In FIGS. **1** to **15** and in the following description, the same reference numerals represent identical or similar elements. In addition, the different elements are not represented to scale so as to enhance clarity of the figures to facilitate understanding thereof. Moreover, the different embodiments and variants do not exclude one another and may be combined together.

(18) In the following description, unless indicated otherwise, the terms «substantially», «about», «generally» and «in the range of» mean «within 10%».

(19) For exclusively illustrative purposes, but without limitation, each of FIGS. **2** to **15** represents an optoelectronic device **10** comprising only one light-emitting diode **11**, that being so for merely illustrative purposes. Indeed, the number of light-emitting diodes **11** is not restrictive per se.

(20) The disclosure firstly covers an optoelectronic device **10**, including at least one three-dimensional light-emitting diode **11**, having micrometric or nanometric dimensions, and whose emitted light rays are focused or made as much directional as possible thanks to the presence of at least one optical lens **14**, an arrangement thereof as well as a method for forming it being described later on.
(21) The disclosure also covers a method for manufacturing an optoelectronic device **10**.
(22) Thanks to the arrangement of three-dimensional light-emitting diodes **11** associated to a micrometric or nanometric optical lens **14**, a particularly targeted application is the provision of an image display screen or in an image projection device.
(23) It is also clear that the embodiments described in this document could find applications in other fields, for example the detection or measurement of electromagnetic radiations or in photovoltaic applications.
(24) The optoelectronic device **10** comprises a substrate **101**, having a support face **101***a*, which is an element common to the different embodiments. The nature of the substrate **101** may be any one and does not brings any limitation herein, any known substrate **101** suited for the implementation of what will be described later on could be considered.
(25) For example, the substrate **101** is constituted by a stacking of a monolithic layer (not represented), a lower electrode layer (not represented) which may be a conductive seed layer and a first electrical insulation layer (not represented). Those skilled in the art could refer, for example to the patent application FR-A1-3053530 for the formation and provision of such a substrate
(26) For example, the support face **101***a* of the substrate **101** is constituted by the exposed face of the aforementioned first electrical insulation layer.
(27) The monolithic layer may be formed in a semiconductor material, whether doped or not, for example silicon or germanium, and more particularly monocrystalline silicon.
(28) It may also be formed by sapphire and even by a III-V semiconductor material, for example GaN. Alternatively, it may comprise a «Silicon On Insulator» or «SOI» type substrate. Alternatively, the monolithic layer may be formed in an electrically-insulating material.
(29) The lower electrode layer is intended to form one or several first lower electrode(s) and may serve as a seed layer for the growth of portions of light-emitting diodes **11**. The lower electrode layer may be continuous or discontinuous. The material composing the lower electrode layer may comprise a nitride, a carbide or a boride of a transition metal from the column IV, V or VI of the periodic table of elements or a combination of these compounds.
(30) The first electrical insulation layer may comprise a first intermediate electrically-insulating layer which covers the lower electrode layer. Besides its electrical insulation properties, it forms a growth mask enabling the growth, for example epitaxial, of doped portions of the light-emitting diodes **11** from through openings delimited by this growth mask locally opening onto the lower electrode layer. The first electrical insulation layer is also involved in ensuring the electrical insulation between the first lower electrodes (not represented) and the second electrodes **21** (shown in FIG. **1**). The first intermediate insulating layer is made of at least one dielectric material such as, for example, a silicon oxide (for example SiO.sub.2 or SiON) or a silicon nitride (for example Si.sub.3Na or SiN), or a silicon oxynitride, an aluminum oxide (for example Al.sub.2O.sub.3) or a hafnium oxide (for example HfO.sub.2). The thickness of the first intermediate insulating layer may be comprised between 5 nm and 1 μm, preferably comprised between 20 nm and 500 nm, for example equal to about 100 nm.
(31) The first electrical insulation layer may include, in addition to the first intermediate insulating layer, a second intermediate insulating layer (not represented) which covers the first lower electrodes and participates in ensuring the electrical insulation between the first lower electrodes and the second upper electrodes **21**. The second intermediate electrically-insulating layer may also cover the growth mask formed by the first intermediate insulating layer. The second intermediate insulating layer may be made of a dielectric material identical to or different from that one of the

growth mask, such as, for example, a silicon oxide (for example SiO.sub.2) or a silicon nitride (for example Si.sub.3Na or SiN), or a silicon oxynitride, an aluminum oxide (for example Al.sub.2O.sub.3) or a hafnium oxide (for example HfO.sub.2). The thickness of the first intermediate insulating layer may be comprised between 5 nm and 1 μm, preferably comprised between 20 nm and 500 nm, for example equal to about 100 nm.
(32) At least one light-emitting diode **11** having a three-dimensional shape comprising a height according to a longitudinal axis **11***b* is formed on the support face **101***a* of the substrate **101**. This longitudinal axis **11***b* extends according to a first direction **112***a* directed transversely to the support face **101***a* of the substrate **101**. In particular, the first direction **112***a* is substantially perpendicular to the general plane in which the substrate **101** extends.
(33) Thus, each light-emitting diode **11** includes a proximal portion directed towards the support face **101***a* of the substrate **101** and, opposite thereto along the first direction **112***a*, a distal portion **11***a* opposite to the proximal portion. In particular, the proximal portion of each light-emitting diode **11** is formed on the support face **101***a* of the substrate, with the provision of an electrical contact with the lower electrode layer. The light-emitting diodes **11** are characterized by a so-called longitudinal first dimension, denoted **112**, measured along the longitudinal axis **11***b* between the distal portion **11***a* and the proximal portion of the light-emitting diode **11**. The light-emitting diodes **11** are also characterized by a so-called transverse second dimension, denoted **113**, corresponding to a dimension considered perpendicularly to the longitudinal axis **11***b*, in particular corresponding to the largest dimension of the section of the light-emitting diode **11** considered in any sectional plane directed perpendicularly to the longitudinal axis **11***b*. In the case where the dimensions of the section evolve along the height of the light-emitting diode **11**, the second transverse dimension considered in this document corresponds to the largest one of the values taken by the largest dimension of all sections over the entire height. The three-dimensional shape taken by each light-emitting diode **11**, with nanometric or micrometric dimensions, may have generally cylindrical wire-like shape, a cone or a cone frustum, or a generally pyramidal shape whereas the section, whether regular or not along the longitudinal axis **11***b* both in its shape as well as in its dimensions, could be circular or not, for example oval or polygonal (for example square, rectangular, triangular, hexagonal). The second transverse dimension **113**, also called minor dimension, which generally corresponds to a thickness of the diode **11** transversely to the first direction **112***a*, is preferably comprised between 5 nm and 20 μm, preferably between 50 nm and 5 μm. The ratio between the first longitudinal dimension **112**, also called major dimension, and the second transverse dimension **113**, is greater than or equal to 1, preferably greater than or equal to 5, and even more preferably greater than or equal to 10. In some embodiments, the second transverse dimension **113** may be smaller than or equal to about 1 μm, preferably comprised between 100 nm and 1 μm, more preferably between 100 nm and 800 nm. In some embodiments, the first longitudinal dimension **112**, which generally corresponds to the height of the light-emitting diode **11** according to the first direction **112***a*, is larger than or equal to 50 nm, preferably comprised between 50 nm and 50 μm, more preferably between 1 μm and 20 μm.
(34) In particular, the disclosure concerns an optoelectronic device **10** wherein both of the first longitudinal dimension **112** and the second transverse dimension **113** of the light-emitting diodes **11** are smaller than or equal to 20 μm.
(35) In general, each light-emitting diode **11** comprises semiconductor elements including a first portion doped according to a first doping type to serve as a P-doped junction, a second portion forming an active portion and a third portion doped according to a second doping type to serve as a N-doped junction. The semiconductor elements of the light-emitting diodes **11** have a three-dimensional shape, which includes the wired shape, the conical or frustoconical shape or the pyramidal shape.
(36) In the description and in the figures, the embodiments are described for the non-limiting particular case where the three-dimensional light-emitting diodes **11** are in the form of a cylindrical

wire whose section is any one, for example circular.
(37) In general, each light-emitting diode **11** is electrically connected to the first lower electrode which is formed in the substrate **101**.
(38) As example, the light-emitting diodes **11** may be made, at least partially, of semiconductor materials from the group IV such as silicon or germanium or mostly including a III-V compound, for example III-N compounds. Examples from the group III comprise gallium, indium or aluminum. Examples of III-N compounds are GaN, AlN, InGaN or AlInGaN. Other elements from the group V may also be used, for example, phosphorus, arsenic or antimony. In general, the elements in the III-V compound may be combined with different molar fractions. It should be set out that the light-emitting diodes **11** may indifferently be formed from semiconductor materials including mostly a II-VI compound. The dopant may be selected, in the case of a III-V compound, from the group comprising a P-type dopant from the group II, for example magnesium, zinc, cadmium or mercury, a P-type dopant from the group IV for example carbon, or a N-type dopant from the group IV, for example silicon, germanium, selenium, sulfur, terbium or tin.
(39) The active portion of the light-emitting diodes **11** corresponds to the layer from which most of the light rays output by the light-emitting diodes **11** is emitted. It may include means for confining the electric charge carriers, such as quantum wells. For example, it is constituted by an alternation of GaN and InGaN layers. The GaN layers may be doped or not. Alternatively, the active portion is constituted by one single InGaN layer.
(40) In general, the light-emitting diodes **11** may be obtained by any technique of a person skilled in the art such as for example: a chemical vapor deposition called «CVD», an atomic layer deposition called «ALD», or a physical vapor deposition called «PVD» or by molecular beam epitaxy «MBE» or by metal organic vapor phase epitaxy «MOVPE» or by metal organic chemical vapor deposition «MOCVD».
(41) As example, each light-emitting diode **11** is electrically connected to the second upper electrode **21** represented in FIG. **1**. The second upper electrode **21** covers at least partially the light-emitting diode **11** so that when a voltage is applied between the first lower electrode and the second upper electrode **21**, light rays are emitted from the active portion of the light-emitting diode **11**. Preferably, the second upper electrode **21** is transparent to these light rays. For example, it may be formed by a transparent conductive oxide «TCO» such as an indium oxide doped with tin («ITO» standing for «Indium Tin Oxide») or a zinc oxide doped for example with aluminum. A person skilled in the art can use any known technique to implement this second upper electrode **21**. The second upper electrode **21** may extend beyond the walls of the light-emitting diode **11**, in particular substantially parallel to the support face **101***a*. In a non-limiting particular example, the second upper electrode **21** may be covered at least partially with a metal to improve conductivity. A person skilled in the art can use any known technique to achieve a contact recovery on the second upper electrode **21** from outside the optoelectronic device **10**.
(42) The light-emitting diodes **11** may also be wrapped at least partially with a passivation layer **114**, as illustrated in FIG. **1**. The passivation layer **114** may be made of an at least partially transparent electrically-insulating material. For example, the passivation layer **114** is made of SiN or SiON. Preferably, the maximum thickness of the passivation layer **114** is comprised between 250 nm and 50 μm and preferably with a thickness of about 1 μm.
(43) In FIG. **1**, the light-emitting diode **11** is covered by the second upper electrode **21**, covered, in turn, by the passivation layer **114**. For clarity, in FIGS. **2** to **15**, the light-emitting diode **11** is represented without the passivation layers **114** and without the second upper electrode **21**. However, the light-emitting diode **11** used in FIGS. **2** to **15** may indifferently be replaced with that one represented in FIG. **1** which is covered by the passivation layer **114** and by the second upper electrode **21**. Similarly, the distal portion **11***a* of the light-emitting diode **11** may be covered, or not, by the second upper electrode **21** and by the passivation layer **114**. Thus, the second transverse dimension **113** may correspond to the transverse dimension of the light-emitting diode **11** with the

presence of the second upper electrode **21** and of the passivation layer **114**.
(44) As illustrated in FIG. **2**, according to a major feature, the optoelectronic device **10** comprises at least one optical lens **14** formed on top of the light-emitting diode **11** according to the first direction **112***a*. The optical lens **14** is formed on the distal portion **11***a* of the light-emitting diode **11** while being generally axially aligned with the longitudinal axis **11***b* of the light-emitting diode **11**. By «formed on top of», it should be understood that the optical lens **14** is either in mechanical contact with the distal portion **11***a* of the light-emitting diode **11**, or located at a distance shorter than 20 μm with respect to the distal portion **11***a* of the light-emitting diode **11** with the interposition of an intermediate material, or not.
(45) By «generally aligned» or «generally axially aligned», it should be herein understood that either the optical axis of the optical lens **14** is collinear within 20 μm with the longitudinal axis **11***b* of the light-emitting diode **11**, or the optical axis of the optical lens **14** is aligned within a 10° angle with respect to the longitudinal axis **11***b* of the light-emitting diode **11**.
(46) The optical lens **14**, which is intended to be crossed by all or part of the light rays emitted by the light-emitting diode **11** in order to transform them as will be described later on, has an outer surface adapted to be crossed by these light rays originating from the light-emitting diode **11** having a convex shape.
(47) In particular, the optical lens **14** may be shaped so that the convex shape of its outer surface is in the form of a surface of revolution whose axis of revolution is substantially aligned with the longitudinal axis **11***b* of the diode **11** and whose generatrix is an arcuate segment, for example a circle arc. The axis of revolution then materializes the optical axis of the optical lens **14**. By «substantially aligned», it should be understood herein that either the axis of revolution of the surface of revolution is collinear within 20 μm with the longitudinal axis **11***b* of the light-emitting diode **11**, or the axis of revolution of the surface of revolution is aligned within a 10° angle with respect to the longitudinal axis **11***b* of the light-emitting diode **11**. It should be recalled that, by mathematical definition, a surface of revolution corresponds to the spatial surface generated by the 360° rotation of the aforementioned generatrix about the axis of revolution.
(48) As example, the surface of revolution has a section, viewed in any sectional plane including the axis of revolution, shaped like a horseshoe arch. This means that the generatrix is then a circle arc covering an angular sector strictly wider than 180°. It goes without saying that while being advantageous for optimizing optical efficiency, this feature is not restrictive.
(49) For example, the optical lens **11** may have a portion truncated according to a plane substantially parallel to the support face **101***a*. In one example, the truncated portion of the optical lens **14** is located on the side opposite to the portion of the optical lens **14** located opposite the distal portion **11***a* of the light-emitting diode **11**.
(50) Advantageously, the optical lens **14** is made of a material with an optical index comprised between 1.4 and 2.2, and more preferably between 1.4 and 1.6.
(51) The lens **14** may be made of borosilicate, glass, silica SiO.sub.2, Al.sub.2O.sub.3, sapphire, polymer, thermoformable polymer, photosensitive resin, plastic, or of a liquid.
(52) Preferably, the optical lens **14** has a shape comprising, totally or partially, of a generally spherical shape, that is to say the shape of all or part of a sphere. By «generally spherical», it should be understood that the diameter of the optical lens **14** considered in any direction varies by less than 10%. Advantageously, this shape allows collecting a maximum of light rays originating from the light-emitting diode **11** thanks to a reduced focal distance.
(53) The optical lens **14** and the distal portion **11***a* of the light-emitting diode **11** may be separated from one another by a distance D**2**, comprised between 0 and 20 μm. Thus, the optical lens **14** and the distal portion **11***a* of the light-emitting diode **11** may be tangential or planar punctual mechanical contact, or be distant from one another by a non-zero distance shorter than 20 μm.
(54) Advantageously, the optical lens **14** allows optically transforming the light rays crossing it, so as to generally focus them or improve the directivity thereof by making them as much parallel as

possible to one another.
(55) In the rest of the text, the terms «optically transform the light rays» should be broadly understood and comprise in particular the actions of focusing or making substantially parallel or ensuring a collimation or improving the directivity of all or part of the light rays emitted by the light-emitting diode **11** that cross the optical lens **14**.
(56) Advantageously, this allows increasing the luminous efficacy of a wire-like shaped three-dimensional light-emitting diode **11** within a determined solid angle as, in the absence of an optical lens **14**, a portion of the light emitted by the light-emitting diode **11** is lost because of its emission outside this solid angle.
(57) Several optical lenses **14** may be arranged in series according to the longitudinal axis **11***b*, for example by combining a spherical first optical lens and a truncated second optical lens superimposed to the first optical lens, for example, so as to make light rays originating from the light-emitting diode **11** parallel thanks to this first optical lens and then focus them using this second optical lens.
(58) The surface of revolution of the optical lens **14** may be shaped so that the surface area of the outer surface, in a section perpendicular to its axis of revolution, has a maximum value at the level of a main plane «P**1**» substantially parallel to the support face **101***a* where the section has a maximum diameter D**1** larger than or equal to the second transverse dimension **113** of the light-emitting diode **11**.
(59) Preferably, the diameter D**1** is comprised between 0.1 μm and 100 μm. An optical lens **14** having a diameter D**1** as described allows recovering a maximum of light rays originating from a light-emitting diode **11**. Advantageously, this allows increasing the luminous efficacy and the overall efficiency.
(60) As illustrated in FIGS. **4** and **5** which represent second and third embodiments, all or part of the space comprised between the portion of the optical lens **14** located opposite the distal portion **11***a* of the light-emitting diode **11** and the distal portion **11***a* of the light-emitting diode **11**, is filled with an optical material **16** with an optical index comprised between 1.4 and 2.2, more preferably between 1.4 and 2, in any case lower than or equal to the optical index of a material of the optical lens **14**.
(61) Throughout the entire text, the terms «optical index» and «refractive index» are equivalent.
(62) For example, the optical material **16** may be formed by SiON or SiN or borosilicate, glass, silica SiO.sub.2, Al.sub.2O.sub.3, sapphire, polymer, thermoformable polymer, photosensitive resin, or plastic. For example, the optical material **16** may be adapted to be etched isotropically through the implementation of a first etching method. In one example, this first etching method may comprise wet or vapor-phase chemical etching, for example, using hydrofluoric acid HF.
(63) The light-emitting diodes **11** are at least partially surrounded by an encapsulation layer **13**. The encapsulation layer **13** may comprise a matrix of an inorganic material, at least partially transparent, within which particles of a dielectric material are possibly scattered. The refractive index of the dielectric material composing the particles is strictly higher than the refractive index of the material composing the matrix. According to one example, the encapsulation layer **13** comprises a layer of silicone, typically of polysiloxane, and further comprises particles of a dielectric material scattered within the matrix. The particles are made of any material type allowing obtaining particles with nanometric dimensions, relatively spherical and having a suitable refractive index. As example, the particles may be made of titanium oxide (TiO.sub.2), zirconium oxide (ZrO.sub.2), zinc sulfide (ZnS), lead sulfide (PbS) or amorphous silicon (Si). The average diameter of a particle corresponds to the diameter of the sphere having the same volume. Preferably, the average diameter of the particles of the dielectric material is comprised between 2 nm and 250 nm. Preferably, the volume concentration of the particles with respect to the total weight of the encapsulation layer **13** is comprised between 1% and 50%. According to another example, the inorganic material is selected from the group comprising silicon oxides of the type SiO.sub.x where

x is a real number strictly greater than 0 and lower than or equal to 2, silicon oxides of the type SiO_yN_z where y is a real number strictly greater than 0 and lower than or equal to 2 and z is strictly greater than 0 and lower than or equal to 0.5, and aluminum oxide (Al_2O_3). The encapsulation layer **13** may be made of an organic material that is at least partially transparent. According to one example, the encapsulation layer **13** is made of polyimide. According to another example, the encapsulation layer **13** is made of an epoxide polymer which further comprises particles of a dielectric material scattered within the matrix. The particles may be made of titanium oxide (TiO_2), zirconium oxide (ZrO_2), zinc sulfide (ZnS), lead sulfide (PbS)) or amorphous silicon (Si). To improve the luminous efficacy of the optoelectronic device **10**, a texturing surface treatment may be applied to the encapsulation layer **13**. In another example, the encapsulation layer **13** is a resin or a photosensitive resin. In another example, the encapsulation layer **13** contains light converters such as photoluminescent pads or quantum pads. The photoluminescent pads are designed so as to absorb and convert at least one portion of the incident light rays, for example blue-colored, coming from the light-emitting diode **11** and to emit exiting light rays of a different color, for example green or red. Advantageously, these photoluminescent pads may form a wavelength filter as described later on. In one example, the encapsulation layer **13** does not cover the distal portion **11***a* of the light-emitting diodes **11** and surrounds only their lateral faces that link the proximal portion and the distal portion **11***a*. Conversely, the encapsulation layer **13** may also covers the distal portion **11***a* of the light-emitting diodes **11**. The thickness of the encapsulation layer **13** considered parallel to the support face **101***a* from the lateral face of the light-emitting diodes **11** is preferably comprised between 500 nm and 50 μm. In an example illustrated in FIG. **5**, the encapsulation layer **13** contains color converters as it allows preventing parasitic wavelengths from being emitted out of the optoelectronic device **10**.

(64) The walls **15** adapted to reflect the light rays emitted by the light-emitting diodes may, for example, be substantially parallel to the longitudinal axis **11***b* of the light-emitting diodes **11** so as to advantageously reflect as much light rays as possible. For example, the walls **15** may be obtained by etching of the encapsulation layer **13** and at least partial filling with a reflective material, such as for example a metal. The walls **15**, together with the encapsulation layer **13**, allow increasing light extraction towards the optical lens **14**. It is still possible to provide for the walls **15** being neither planar nor vertical. Preferably, the walls **15** have a height, considered according to the longitudinal axis **11***b* from the support face **101***a* of the substrate **11**, larger than the height of the encapsulation layer **13** measured the same way. Thus, a maximum of the light rays emitted by the light-emitting diode **11** could be reflected and/or guided towards the optical lens **14**.

(65) As it has already been indicated, the disclosure also concerns a method for manufacturing an optoelectronic device **10**, in particular the optoelectronic device **10** as described before.

(66) In general, the method for manufacturing the optoelectronic device **10** comprises the following steps: a) formation of a substrate **101** delimiting a support face **101***a*, b) formation of at least one light-emitting diode **11** having a three-dimensional shape comprising a height along a longitudinal axis **11***b* extending according to a first direction **112***a* directed transversely to the support face **101***a*, and having a first longitudinal dimension **112** considered according to the longitudinal axis **11***b* between a proximal portion of the light-emitting diode **11** directed towards the support face **101***a* and a distal portion **11***a* of the light-emitting diode **11** opposite, along the first direction **112***a*, to said proximal portion, and at least one second transverse dimension **113** corresponding to a dimension of said three-dimensional shape considered perpendicularly to the longitudinal axis **11***b*.

(67) Each of the first longitudinal dimension **112** and the second transverse dimension **113** is smaller than or equal to substantially 20 μm.

(68) The manufacturing method also comprises a step i) of forming at least one optical lens **14** on top of the distal portion **11***a* of the light-emitting diode **11** according to the first direction **112***a* at a location such that the optical lens **14** is generally axially aligned with the longitudinal axis **11***b* of the light-emitting diode **11**, the formed optical lens **14** being adapted to transform the light rays

emitted by the light-emitting diode **11** that cross the optical lens **14** and having an outer surface adapted to be crossed by said light rays having a convex shape.
(69) FIGS. **6** to **11** and **14** represent the steps of a first example of the manufacturing method.
(70) To form the optical lens **14**, the manufacturing method may comprise the following steps: c) formation of an encapsulation layer **13** surrounding all or part of the light-emitting diode **11**. The encapsulation layer **13** as well as the distal portion **11***a* of the light-emitting diode **11** may also be optionally flattened and clipped through a mechanical-chemical polishing step; d) formation, on the exposed face of the intermediate structure obtained after the step of forming the encapsulation layer **13**, of a first layer of an optical material **16** with an optical index comprised between 1.4 and 2.2 and lower than or equal to the optical index of the optical lens **14**. This first optical material layer **16** is adapted to be isotropically etched through the implementation of first etching method as described before. The thickness of the optical material layer **16** is comprised between 1 and 50 μm; e) formation, on the exposed surface of the intermediate structure obtained after the step of forming the first optical material layer **16**, of a hard mask layer **17** adapted to not be etched by the implementation of the first etching method. This difference in behavior with regards to the first etching method with respect to the optical material layer **16** advantageously allows obtaining an overhanging structure. Preferably, the hard mask layer **17** as a thickness comprised between 50 nm and 5 μm. The hard mask layer **17** may for example be made of SiN, SiON or SiOCH, f) etching of a first opening **171**, through the implementation of a second etching method, through the hard mask layer **17** obtained at step e), the first opening **171** having dimensions and shapes substantially identical to those of a section of the light-emitting diode **11** according to a plane parallel to the support face **101***a*, the first opening **171** of the hard mask layer **17** being located on top of the light-emitting diode **11** in the continuation of the longitudinal axis **11***b*. For example, the second etching method is of reactive ion etching «RIE» type or of the plasma etching type; g) etching of at least one cavity **19**, through the implementation of the first etching method at the level of the first opening **171** obtained at step f), across all or part of the thickness of the first optical material layer **16**, the cavity **19** obtained in this manner having a concave shape complementary to the convex surface of the optical lens **14** and being suited for the subsequent formation of the optical lens **14** at step i), directly in the cavity **19**; h) removal of the hard mask layer **17**.
(71) For example, this step h) may be carried out by mechanical-chemical polishing or using a selective wet-chemistry.
(72) Part of the previously-described step i) includes arranging the optical lens **14** in the concave cavity **19**. By «arrangement», it should be understood the «formation in situ» or the «formation ex situ with placement».
(73) In one example, steps a), b) and i) are successive with or without other intermediate steps.
(74) In a second example illustrated in FIGS. **12** and **13**, the optical lens **14** is formed in-situ thanks to the implementation of the successive following three substeps: a first substep i1) including forming a polymer layer **18** over a free surface obtained after the hard mask layer **17** has been removed, the polymer layer **18** filling at least partially the concave cavity **19**; to obtain a lens **14** having at least one spherical portion, the cavity **19** is preferably filled with the polymer layer **18**; a second substep i2) including etching a portion to be removed **22** of the polymer layer **18**, leaving at least one residual portion **20** of the polymer layer **18** limited right above the etched portion of the first optical material layer **16**; a third substep i3) including annealing the polymer residual portion **20** where the annealing temperature may be comprised between 100 and 300° C., advantageously, the annealing enables the polymer to pass through a viscous phase so as to advantageously obtain a spherical or curved shape by surface tension forces.
(75) In a third example of the manufacturing method illustrated in FIG. **15**, the latter comprises an additional step j) which includes forming at least one reflective wall **15**, for example by etching of the encapsulation layer **13** and by an at least partial filling with metal. Any other technique known to those skilled in the art may be used. Advantageously, the wall **15** extends from the support face

101*a* of the substrate **101** and substantially parallel to the longitudinal axis **11***b*. Advantageously, the walls **15** surrounding the light-emitting diode **11** allows increasing the luminous efficacy by simple or multiple reflections towards the optical lens **14** of the light rays emitted by the light-emitting diode **11**. Preferably, the distance between the light-emitting diode **11** and the walls **15** is larger than or equal to 500 nm.

Claims

1. An optoelectronic device comprising: a substrate delimiting a support face and at least one light-emitting diode formed on the support face, said at least one light-emitting diode having a three-dimensional shape comprising a height according to a longitudinal axis extending according to a first direction directed transversely to the support face and having a first longitudinal dimension considered according to the longitudinal axis between a proximal portion of the light-emitting diode directed towards the support face and a distal portion of the light-emitting diode opposite, along the first direction, to said proximal portion, and at least one second transverse dimension corresponding to a dimension of said three-dimensional shape considered perpendicularly to the longitudinal axis, wherein each of the first longitudinal dimension and the second transverse dimension is smaller than or equal to about 20 μm, the optoelectronic device includes an optical lens adapted to transform the light rays emitted by the light-emitting diode that cross said optical lens, said optical lens being formed above the distal portion of the light-emitting diode according to the first direction and having an optical axis generally axially aligned with the longitudinal axis of the light-emitting diode and having an outer surface facing the distal portion of said light-emitting diode, adapted to be crossed by at least one portion of said light rays to be transformed, said outer surface having a convex shape.
2. The optoelectronic device according to claim 1, wherein the convex shape of the outer surface of the optical lens is in the form of a surface of revolution whose axis of revolution is substantially aligned with the longitudinal axis of the light-emitting diode and whose generatrix is an arcuate segment.
3. The optoelectronic device according to claim 2, wherein the surface of revolution has a section, viewed in any sectional plane including the axis of revolution, having a horseshoe arch shape.
4. The optoelectronic device according to claim 2, wherein the surface of revolution features the shape of all or part of a sphere.
5. The optoelectronic device according to claim 2, wherein the surface of revolution is shaped so that the surface area of the section of the outer surface in a section perpendicularly to its axis of revolution has a maximum value at the level of a main plane substantially parallel to the support face where the section has a maximum diameter larger than or equal to the second transverse dimension of the light-emitting diode above which the optical lens having said maximum diameter is formed.
6. The optoelectronic device according to claim 5, wherein the maximum diameter is comprised between 0.1 μm and 20 μm.
7. The optoelectronic device according to claim 1, wherein the optical lens is made of a material having a refractive index comprised between 1.4 and 2.2.
8. The optoelectronic device according to claim 1, wherein on a side opposite to the distal portion of the light-emitting diode according to the first direction, the optical lens is truncated according to a plane substantially parallel to the support face.
9. The optoelectronic device according to claim 1, wherein at least one portion of the light-emitting diode comprises a passivation layer.
10. The optoelectronic device according to claim 1, wherein the optical lens and the distal portion of the light-emitting diode are in mechanical contact.
11. The optoelectronic device according to claim 1, wherein at least one portion of the optical lens

and the distal portion of the light-emitting diode are separated from one another with interposition of a distance comprised between 5 nm and 20 μm.

12. The optoelectronic device according to claim 1, wherein an optical material, with a refractive index comprised between 1.4 and 2.2 and lower than or equal to a refractive index of the optical lens, fills all or part of the space comprised between the distal portion of the light-emitting diode and the portion of the optical lens turned towards the distal portion of the light-emitting diode.

13. The optoelectronic device according to claim 1, wherein all or part of the light-emitting diode is surrounded by an encapsulation material.

14. The optoelectronic device according to claim 13, wherein, when viewed perpendicularly to the longitudinal axis, the encapsulation material is surrounded by at least one wall adapted to reflect the light rays emitted by the light-emitting diode.

15. A manufacturing method for manufacturing an optoelectronic device, the method including the following successive steps: a) formation of a substrate delimiting a support face, b) formation of at least one light-emitting diode having a three-dimensional shape comprising a height according to a longitudinal axis extending according to a first direction directed transversely to the support face and having a first longitudinal dimension considered according to the longitudinal axis between a proximal portion of the light-emitting diode directed towards the support face and a distal portion of the light-emitting diode opposite, along the first direction, to said proximal portion, and at least one second transverse dimension corresponding to a dimension of said three-dimensional shape considered perpendicularly to the longitudinal axis, wherein each of the first longitudinal dimension and the second transverse dimension is smaller than or equal to substantially 20 μm, and i) formation of one optical lens above the distal portion of the light-emitting diode according to the first direction so that the optical axis of the optical lens is generally axially aligned with the longitudinal axis of the light-emitting diode, the formed optical lens being adapted to transform the light rays emitted by the light-emitting diode that cross the optical lens and having an outer surface facing the distal portion of said light-emitting diode, adapted to be crossed by said light rays to be transformed, said outer surface having a convex shape.

16. The manufacturing method according to claim 15, including the following steps, implemented between step b) and step i): c) formation of an encapsulation material surrounding all or part of the light-emitting diode, d) formation, over the exposed face of the intermediate structure obtained at step c), of a first layer of an optical material, with a refractive index comprised between 1.4 and 2.2 and lower than or equal to a refractive index of the optical lens, said first optical material layer being adapted to be etched isotropically through the implementation of a first etching method, e) formation, over the free surface of the intermediate structure obtained after step d), of a hard mask layer adapted to not be etched during the use of the first etching method, f) etching of a first opening, through the implementation of a second etching method, through the hard mask layer obtained at step e), the first opening corresponding to a section of the light-emitting diode according to a plane parallel to the support face, said first opening of the hard mask layer being located above the light-emitting diode in the continuation of the longitudinal axis, g) etching of at least one cavity, through the implementation of the first etching method at the level of the first opening obtained at step f), across all or part of the thickness of the first optical material layer, the cavity obtained in this manner having a concave shape complementary to the convex surface of the optical lens and being suited for the subsequent formation of the optical lens at step i) directly in the cavity, and h) removal of the hard mask layer.

17. The manufacturing method according to claim 16, wherein step i) includes the following steps: i1) formation of a polymer layer over the exposed surface of the intermediate structure obtained after step h), the polymer layer filling all or part of the cavity, i2) etching of the polymer layer located at the level of at least one portion to be removed, allowing preserving at least one residual portion of the polymer layer located right above the light-emitting diode in the continuation of the longitudinal axis, and i3) annealing of said at least one residual portion, annealing being carried out

in conditions allowing making the polymer constituting said at least one residual portion pass through a viscous state.

18. The manufacturing method according to claim 15, including a step j) of forming at least one wall adapted to reflect the light radiations emitted by the light-emitting diode, the wall extending from the support face substantially parallel to the longitudinal axis.